HUARAVITO LTEMS

Long-Term Ecological Monitoring Site

Date: February 25, 2026

Location: (10.104583, -84.564639)

Piedades Sur, San Ramón, Costa Rica

Reserva Huaravito Long-Term Ecological Monitoring Site (LTEMS)

Technical Note

Comparative Thermodynamic Analysis:

Degraded Anthropogenic Site (Tajo) vs. Old-Growth Forest Core (Toyopán)

- Subject: Empirical Documentation of the Toyopán Holobiont Thermoregulatory Capacity.

- Validated by Dual-Method Protocol (Digital Array + Analog Mercury Thermometer)

§ 1 · Metadata

- **OBSERVER:** Randall Santamaría**,** Custodian & Director, Huaravito LTEMS
- **DATE:** February 27, 2026
- **TIMEFRAME:** 11:55 AM – 12:41 PM (Peak Solar Radiation Window)
- **DURATION:** 54 minutes
- **CONDITIONS:** High radiation · Zero wind · Minimal cloud cover (High Heat Index)
- **INSTRUMENTATION:** Digital Sensory Array + Analog Mercury Thermometer (cross-validation protocol)
- **METHOD:** Linear transect · continuous sampling · fixed-station termini
- **ECOSYSTEM:** Premontane moist forest / anthropogenic degraded matrix

§ 2 · Dataset: Linear Transect Temperatures

Time	Station / Biotic Feature	Temp (°C)	Field Observation
11:56 AM	Tajo Floor (Dry Clay Crater)	34.5	Maximum anthropogenic heat. Zero biotic buffering. Bare compacted clay substrate.
12:00 PM	Tajo Slope (Interior)	31.9	Exposed soil; lateral radiant heat from crater walls.
12:01 PM	Access Road to Tajo	29.3	Transition zone; open canopy; surface albedo near bare ground.
12:03 PM	Toyopán Forest Entrance	27.9	Initial canopy buffering; first epiphyte-bearing stems detected.
12:05 PM	Guayabón Tree (Buttress Roots)	26.2	Micro-niche cooling; high epiphyte biomass (>70% cover).
12:11 PM	Mature Forest Interior	24.9	Stable shade; closed multi-strata canopy; high humidity.
12:28 PM	Hauya elegans (Giant Specimen)	24.8	Biogenic stability confirmed. Moss-angiosperm continuum on bole.
12:41 PM	Toyopán Sacred Forest Core	24.5	Minimum biological temperature. Mercury thermometer cross-validation performed.

THERMAL GRADIENT SUMMARY: 34.5°C → 24.5°C over ~150 m
ΔT = **10.0°C** (*Maximum thermal delta confirmed*)

§ 3 · Thermodynamic Synthesis

Maximum Thermal Delta confirmed between the degraded anthropogenic landscape (Tajo, **34.5°C**) and the core of the Toyopán Holobiont (**24.5°C**). This differential represents empirical evidence of active biogenic thermoregulation at ecosystem scale, validated by dual-method instrumentation.

- Biogenic Cooling Efficiency

The forest core maintained a near-steady-state temperature of **24.5°C** during peak solar incidence, representing a suppression of **10.0°C** relative to the degraded matrix.

- Analog Validation

The analog mercury thermometer readings confirm the thermal differential as a physical microclimatic reality, excluding digital sensor drift or calibration error.

- Homeostatic Resilience

Temperatures declined monotonically from entrance (27.9°C) to core (24.5°C), with final stations holding within a 0.4°C variance band under peak radiation.

- Thermal Refuge Function

The Toyopán core functions as a thermal refuge for heat-sensitive taxa including Hauya elegans and associated epiphytic and invertebrate communities.

§ 4 · Custodian's Field Note

The 54-minute transect under extreme solar radiation and zero wind highlighted the thermal contrast between degraded and forested systems. The physical heat stress experienced outside the forest boundary and the immediate relief upon entering the Toyopán core provide somatic corroboration of the instrumental dataset.

This qualitative observation is archived as a primary field record.

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